



Instructions For Use

TRM-IFU

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Trichrome Stain Kit

(Modified Masson's)

Description and Principle

The Trichrome Stain Kit (Modified Masson's) is intended for use in the histological visualization of collagenous connective tissue fibers in tissue sections. This kit may be used on formalin-fixed, paraffin-embedded sections.

This kit utilizes 3 dye solutions used in sequence along with an Iron hematoxylin solution that resists decolorization by subsequent acidic stain solutions. Sections are first mordanted in heated Bouin's Fluid which acts to intensify the subsequent trichrome stains. Biebrich Scarlet / Acid Fuchsin Solution stains all acidophilic tissue elements red. Phosphomolybdic and phosphotungstic acids displace the red dye out of collagen fibers leaving them colorless and receptive to staining by aniline blue.

Expected Results

Collagen:	Blue
Muscle Fibers:	Red
Nuclei:	Dark Red to Black/Blue

Kit Contents

1. Bouin's Fluid
2. Weigert's Iron, Hematoxylin (A)
3. Weigert's Iron, Hematoxylin (B)
4. Biebrich Scarlet / Acid Fuchsin Solution
5. Phosphomolybdic/Phosphotungstic Acid Solution
6. Aniline Blue Solution
7. Acetic Acid Solution (1%)

Storage

- 18-25°C
- 18-25°C
- 18-25°C
- 18-25°C
- 18-25°C
- 18-25°C
- 18-25°C

Suggested Controls (not provided)

Lung, Liver, Colon, Stomach.

Uses/Limitations

For In-Vitro Diagnostic use only.
Do not use if reagents become cloudy or precipitate
Do not use past expiration date.
Use caution when handling reagents.
Non-Sterile
Intended for FFPE sections cut at 5-10µm.
This procedure has not been optimized for frozen sections.
Frozen sections may require protocol modification.

Storage

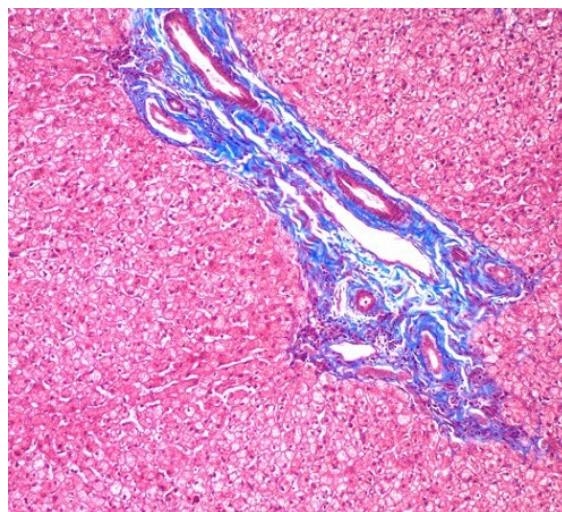
Store kit and all components at room temperature (18-25°C).

Safety and Precautions

Please see current Safety Data Sheets (SDS) for this product and components GHS classification, pictograms, and full hazard/precautionary statements.

Important Notes:

1. If final background is dark or dull red/blue, it may be caused by excess gray background from Weigert's Iron Hematoxylin. If this is the case, reduce staining time of Weigert's Iron Hematoxylin for future slides (step 6). A standard Acid Alcohol solution (not provided) may also be used to remove excess gray background immediately after step 6 as well.



Human Liver stained with Trichrome Stain Kit (Modified Masson's)
viewed at 100X magnification

2. Differentiating strength of Phosphomolybdic/Phosphotungstic Acid Solution is greatly increased by mixing 1:1 with ethanol (not provided). This modification may allow better visualization of small collagen fibers than the traditional method, but will remove red stain much more quickly so should be done with caution:

For example, mix:

-1ml of ethanol

-1ml of Phosphomolybdic/ Phosphotungstic Acid Solution

Replace step 11 by applying 1:1 working solution to tissue for 2-6 minutes. Check slide microscopically for proper differentiation and repeat if needed. Rinse well with deionized water after differentiation.

Procedure:

1. Deparaffinize sections if necessary and hydrate to distilled water.
2. Preheat Bouin's Fluid in a water bath to 56° - 64° centigrade in a fume hood or very well-ventilated area.
3. Place slides in preheated Bouin's Fluid for 60 minutes followed by a 10-minute cooling period.
4. Rinse slide in tap water until section is completely clear.
5. Rinse once in distilled water.
6. Mix equal parts of Weigert's (A) and Weigert's (B) and stain slide with working Weigert's Iron Hematoxylin for 2-4 minutes. Stain is alcoholic and prone to evaporation – monitor and add stain as necessary to ensure stain does not dry on slide. Dried stain may result in excess grey background.
7. Rinse slide in running tap water for 2 minutes.
8. Rinse slide in distilled water.
9. Apply Biebrich Scarlet / Acid Fuchsin Solution to slide for 5-10 minutes.
10. Rinse slide in distilled water.

11. Differentiate in Phosphomolybdic/Phosphotungstic Acid Solution for 10-15 minutes. *Note: See alternative modification in Important Note #2 above.*
12. Without rinsing, apply Aniline Blue Solution to slide for 5-10 minutes.
13. Rinse slide in distilled water.
14. Apply Acetic Acid Solution (1%) to slide for 3-5 minutes.
15. Dehydrate very quickly in 2 changes of 95% Alcohol, followed by 2 changes of Absolute Alcohol.
16. Clear in Xylene or Xylene Substitute, and mount in synthetic resin.

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